

S.Y.B.Sc.(Data Science)

Sem- IV

Subject: Data Structure-II

Unit1 : Tree

By,

Mrs.Reshma Masurekar

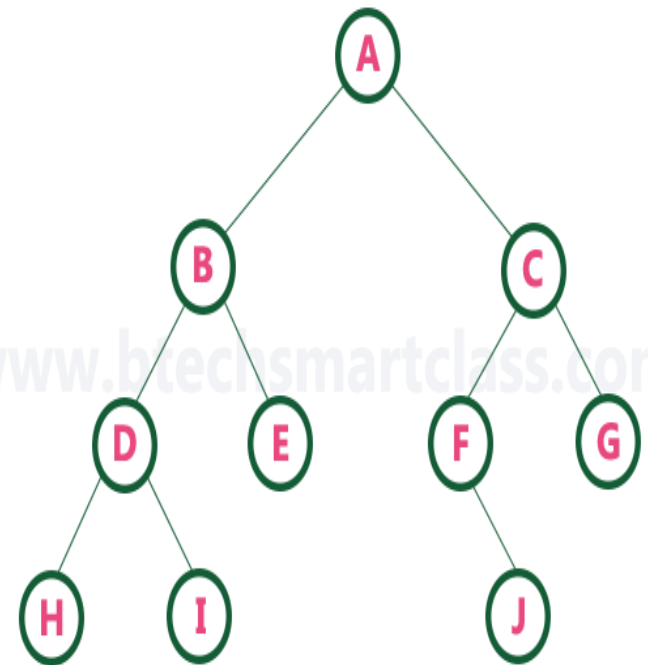
***Dr. D. Y. Patil Arts, Commerce and Science College,
Pimpri, Pune 18***

Contents

- Types of Binary trees - Binary tree, skewed tree, strictly binary tree, full binary tree, complete binary tree, expression tree, binary search tree, Heap

Binary Tree

- In a normal tree, every node can have any number of children.
- A binary tree is a special type of tree data structure in which every node can have a **maximum of 2 children**. One is known as a **left child** and the other is known as **right child**.
- **A tree in which every node can have a maximum of two children is called Binary Tree.**
- In a binary tree, every node can have either 0 children or 1 child or 2 children but not more than 2 children.



Types of Binary trees

- - Binary tree, skewed tree, **strictly binary** tree, full binary tree, **complete binary tree**, expression tree, binary search tree, Heap

Strictly Binary Tree

Def: A full binary tree is a tree in which every node other than the leaves has two children.

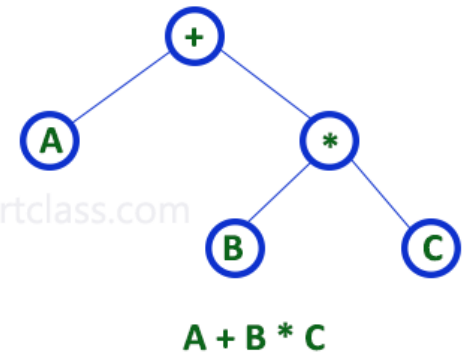
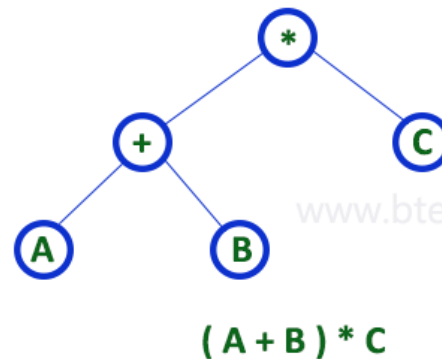
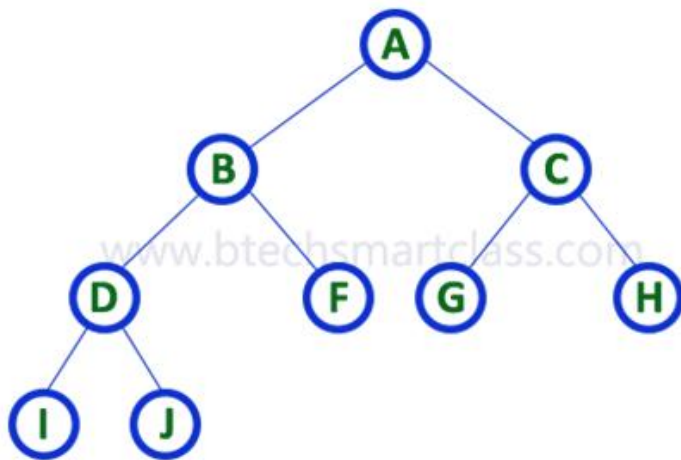
A binary tree in which every internal node has **either two or zero number of children** is called Strictly Binary Tree.

Degree of every nonleaf node is always be 2.

No. of leaf nodes=no.of internal nodes+1

Strictly binary tree is also called as **Full Binary Tree** or **Proper Binary Tree** or **2-Tree**.

Full binary tree is used to represent mathematical expressions.



- **Full Binary Tree Theorems**

- Let,

i = the number of internal nodes

n = be the total number of nodes

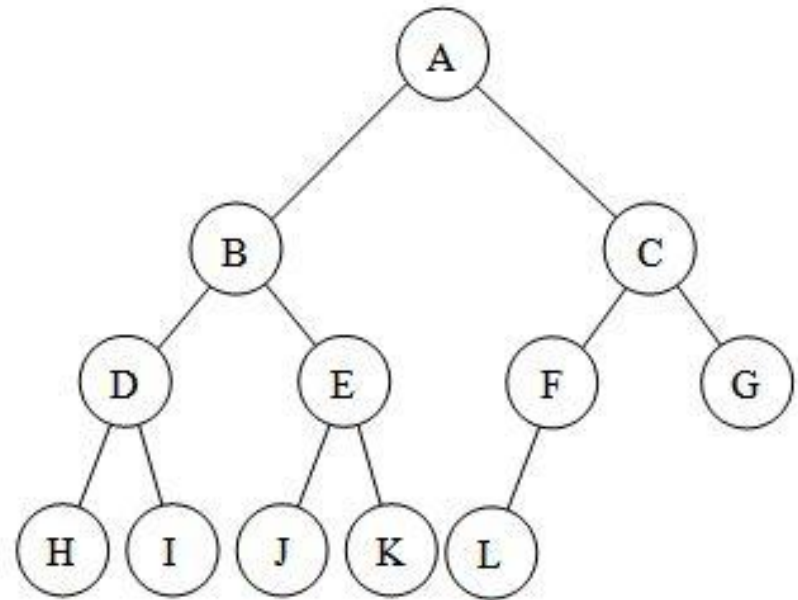
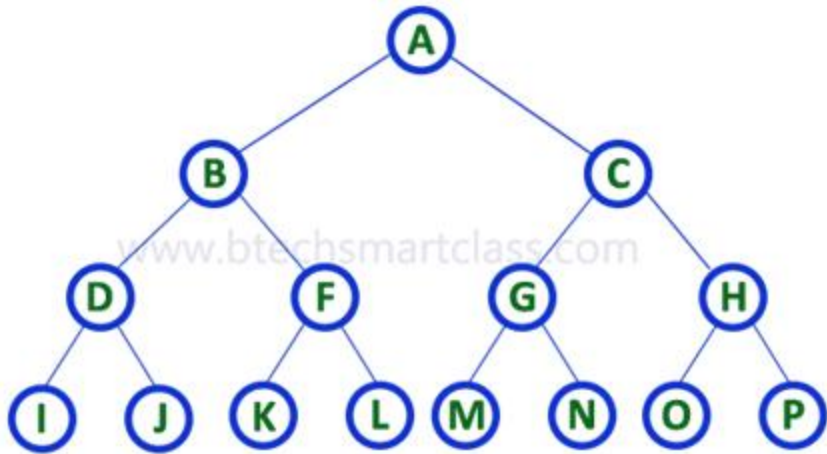
l = number of leaves

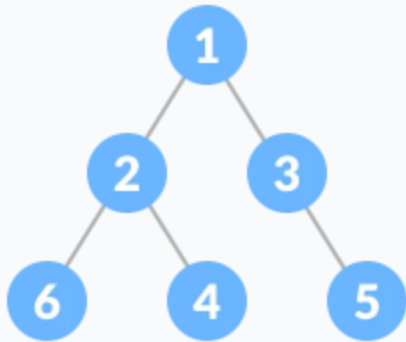
λ = number of levels

- The number of leaves is $i + 1$.
- The total number of nodes is $2i + 1$.
- The number of internal nodes is $(n - 1) / 2$.
- The number of leaves is $(n + 1) / 2$.
- The total number of nodes is $2l - 1$.
- The number of internal nodes is $l - 1$.
- The number of leaves is at most $2^{\lambda - 1}$.

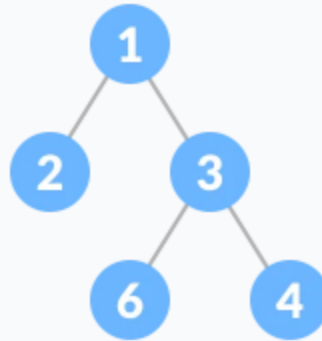
Complete Binary Tree

Def: A binary tree is said to be a complete binary tree **when all the levels are completely filled except the last level, which is filled from the left.**

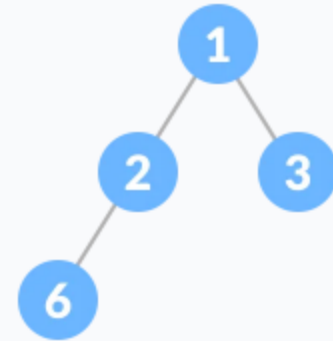




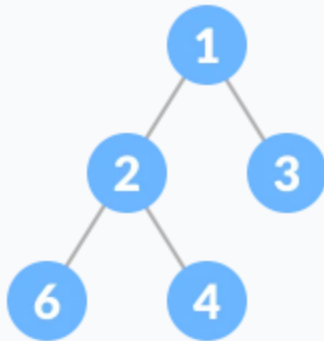
✗ Full Binary Tree
✗ Complete Binary Tree



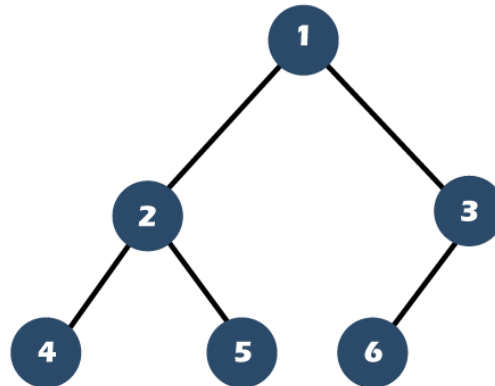
✓ Full Binary Tree
✗ Complete Binary Tree



✗ Full Binary Tree
✓ Complete Binary Tree



✓ Full Binary Tree
✓ Complete Binary Tree



✗ Full Binary Tree
✓ Complete Binary Tree

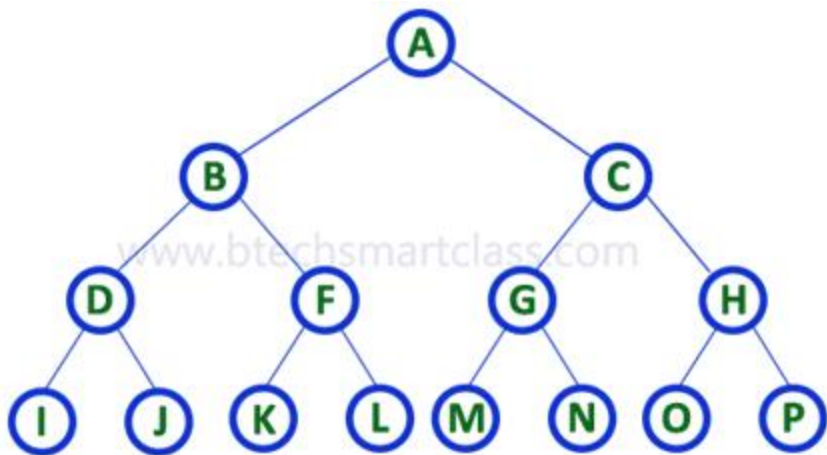
Perfect Binary Tree

Def: A perfect binary tree is a type of binary tree in which every internal node has **exactly two child nodes** and **all the leaf nodes are at the same level**.

All the internal nodes have a degree of 2.

Recursively, a perfect binary tree can be defined as:

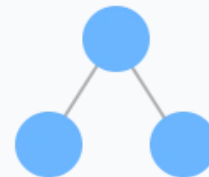
- If a single node has no children, it is a perfect binary tree of height $h = 0$,
- If a node has $h > 0$, it is a perfect binary tree if both of its subtrees are of height $h - 1$ and are non-overlapping.



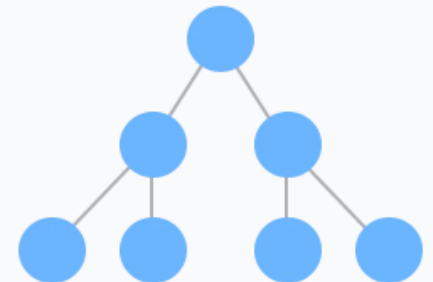
tree-1



tree-2



tree-3



Perfect Binary Tree (Recursive Representation)

- **Perfect Binary Tree Theorems**

- A perfect binary tree of height h has $2^{h+1} - 1$ nodes.
- A perfect binary tree with n nodes has height $\log(n + 1) - 1 = \Theta(\ln(n))$.
- A perfect binary tree of height h has 2^h leaf nodes.
- The average depth of a node in a perfect binary tree is $\Theta(\ln(n))$.

- **Properties of a perfect binary tree:**

- The number of leaf nodes in a perfect binary tree with internal nodes is equal to the number of internal nodes + 1
- The number of nodes in a Perfect Binary Tree of height h is $2^{h+1} - 1$
- All internal nodes of a perfect binary tree have a degree of 2
- All leaf nodes of a perfect binary tree have a degree of 0
- Perfect binary trees are always symmetrical and balanced

- The complete binary tree is similar to the full binary tree except for the two differences which are given below:
 - The filling of the leaf node must start from the leftmost side.
 - It is not mandatory that the last leaf node must have the right sibling.

skewed binary tree

- A binary tree can be called a skewed binary tree if all nodes have one child or no child at all.
- They can be of two types:
 - Left Skewed Binary Tree
 - Right Skewed Binary Tree

- **Left Skewed Binary Tree:** If all nodes are having a left child or no child at all then it can be called a left skewed binary tree. In this tree all children at right side remain null. It is a left side dominated tree.
- **Right Skewed Binary Tree:** If all nodes are having a right child or no child at all then it can be called a right skewed binary tree. In this tree all children at left side remain null. It is a right side dominated tree.

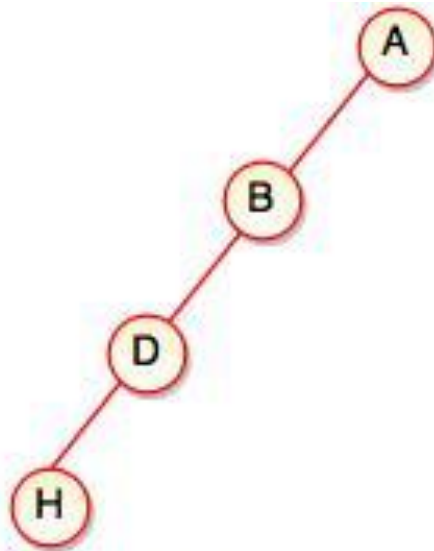


Fig. Left Skewed
Binary Tree

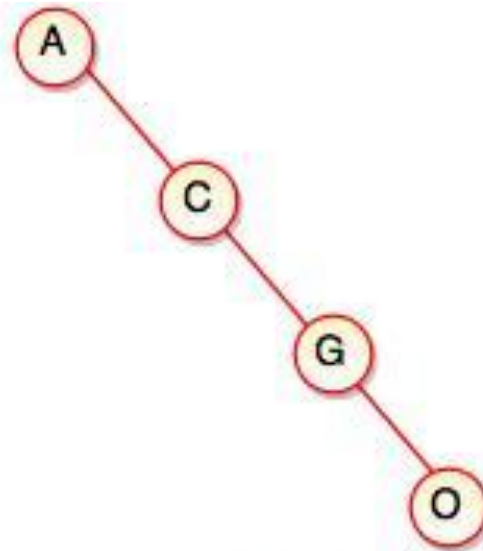
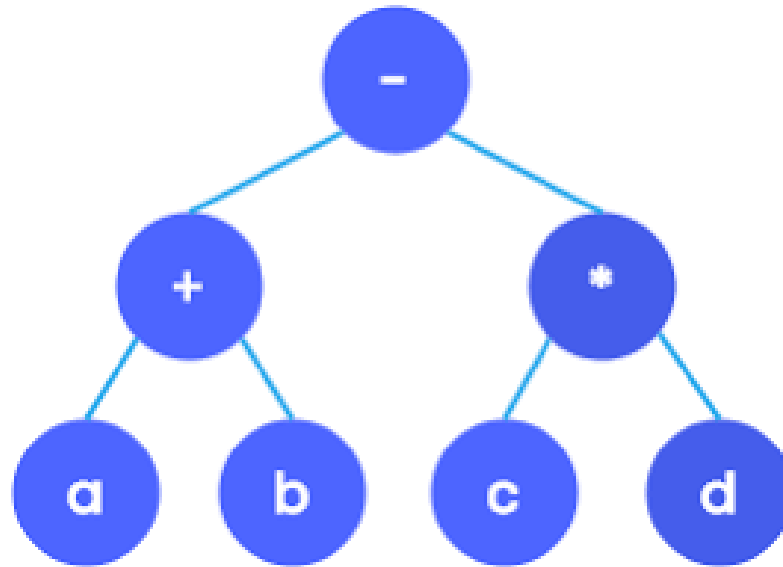


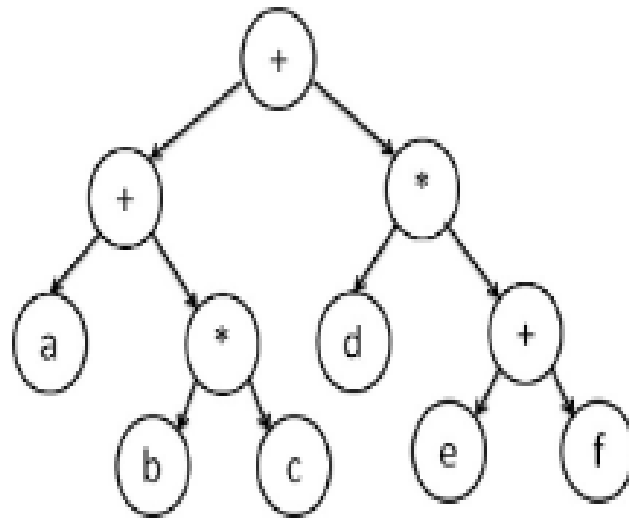
Fig. Right Skewed
Binary Tree

Expression tree

- The expression tree is a binary tree in which each internal node corresponds to the operator and each leaf node corresponds to the operand.
- Example



$a + (b * c) + d * (e + f)$

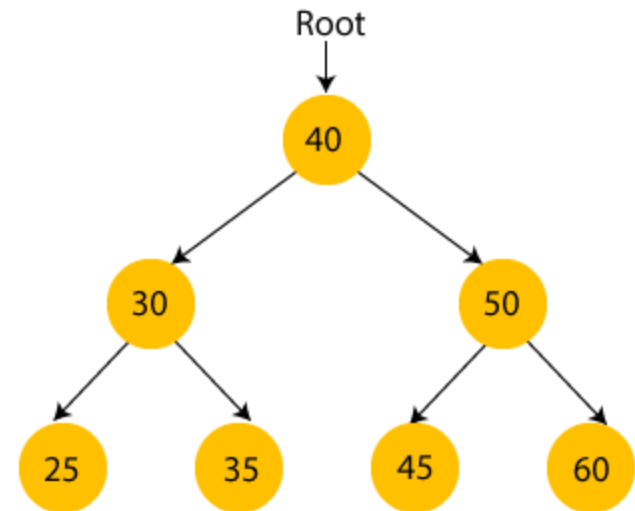
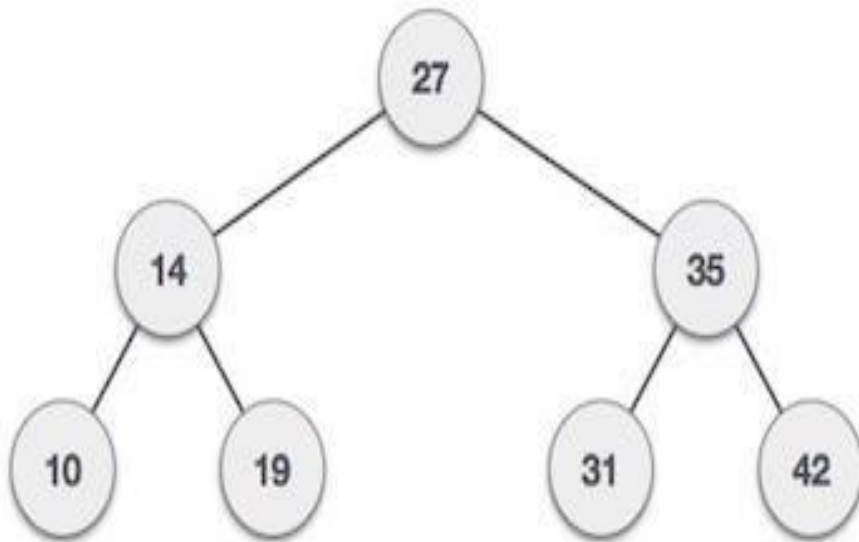


Expression Tree

Binary Search Tree

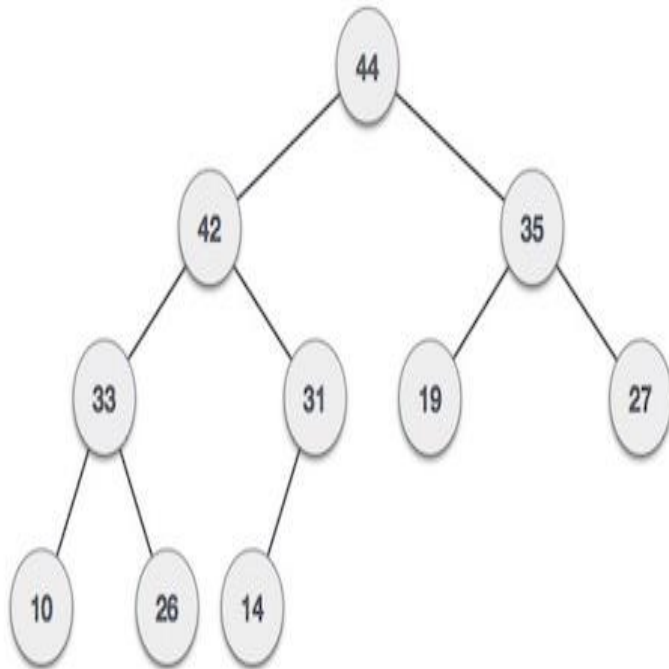
- A binary search tree is a binary tree in which the nodes are arranged according to their values.
- Binary search tree
 - All nodes of **left subtree** are **less** than the root node.
 - All nodes of **right subtree** are **greater** than the root node.
 - Both subtrees of each node are also BSTs i.e. they have the above two properties.
- It is called a search tree because it can be used to search for the presence of a number in **$O(\log(n))$** time.

Binary Search Tree Example

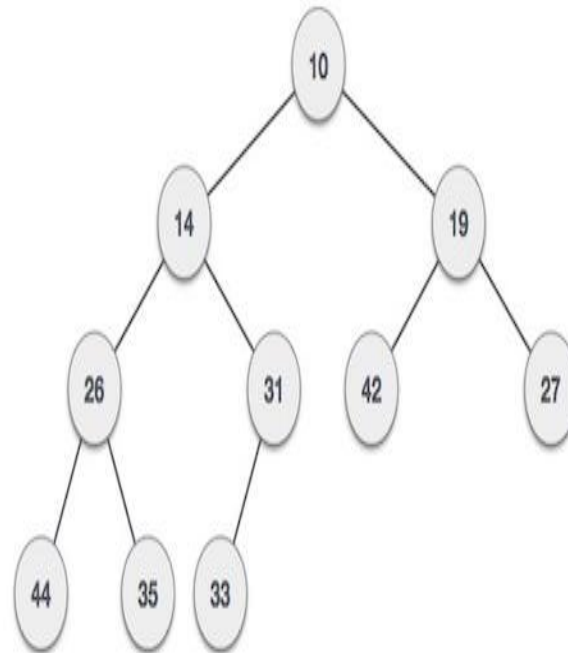


Heap tree

- A heap is a **complete binary tree**, and the binary tree is a tree in which the node can have **utmost two children** that satisfies the heap property, where any given node is
 - always greater than its child node/s and the key of the root node is the largest among all other nodes. This property is also called **max heap**.
 - always smaller than the child node/s and the key of the root node is the smallest among all other nodes. This property is also called **min heap**.



Max heap



Min heap